

NLP-Based Plagiarism Detection System

Using **BERT (Sentence-BERT)** to detect semantic similarity and paraphrased plagiarism — beyond simple keyword matching.

NLP · BERT · SEMANTIC SIMILARITY

Introduction

Traditional plagiarism detectors rely on exact word matching — they fail when text is paraphrased. This system uses **Sentence-BERT embeddings** to understand the *meaning* of text, not just its surface form.

Compare Two Documents

Sentence-by-sentence semantic comparison of uploaded PDFs.

Single Document Scan

Detect external plagiarism by comparing against a reference corpus.

Paraphrase Detection

Identify rewritten content that preserves original meaning.

PDF Report Output

Generate highlighted, percentage-scored plagiarism reports.

Problem Statement

The Core Challenge

Keyword-based systems cannot detect **semantic plagiarism** — when ideas are copied but words are changed. Paraphrasing tools make this trivial to bypass.

✗ Exact Match Only

Misses all reworded or synonym-swapped content.

✗ No Context Awareness

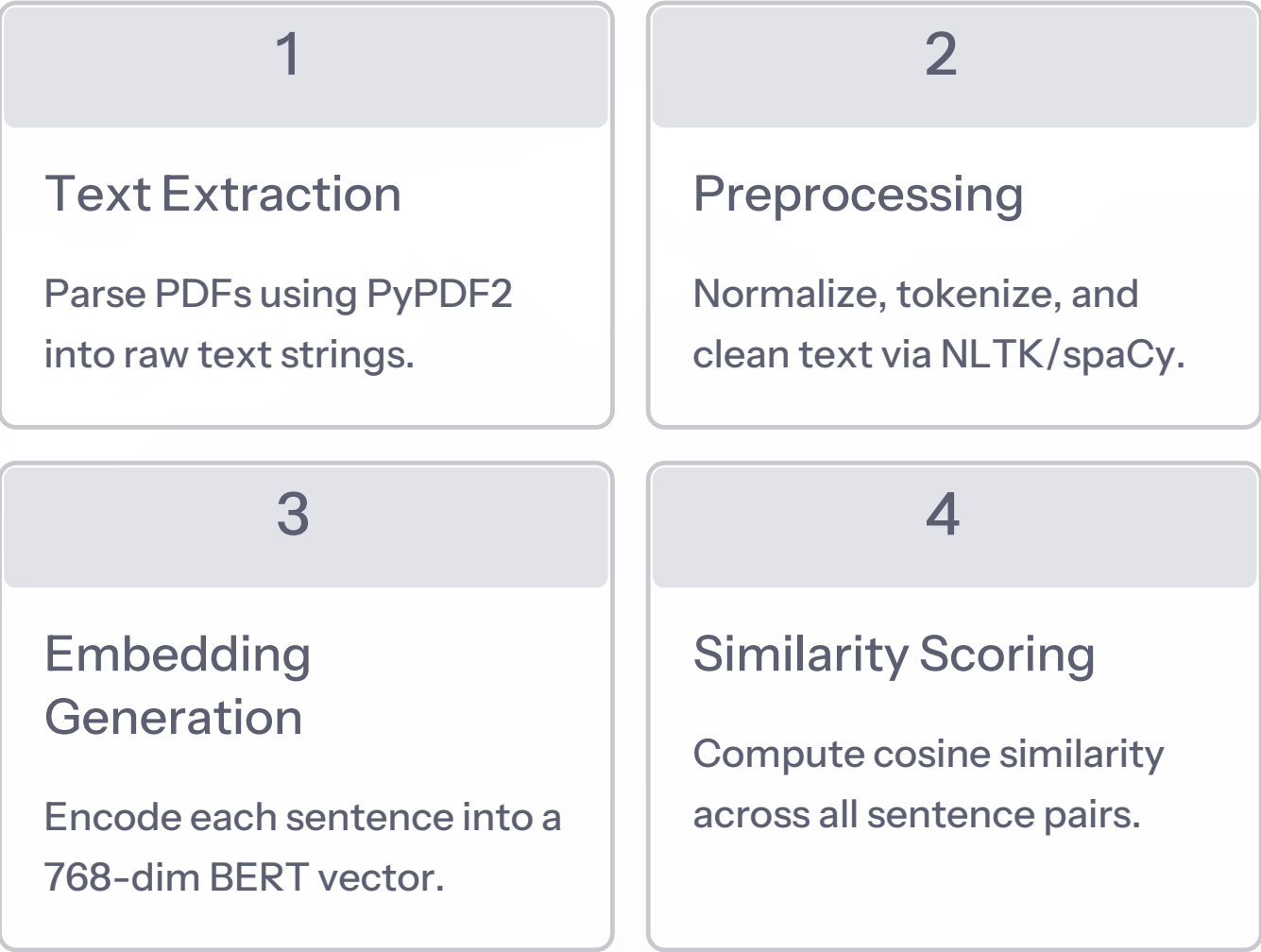
Cannot distinguish similar words from similar meaning.

✗ High False Negatives

Paraphrased plagiarism goes undetected in academic submissions.

Methodology

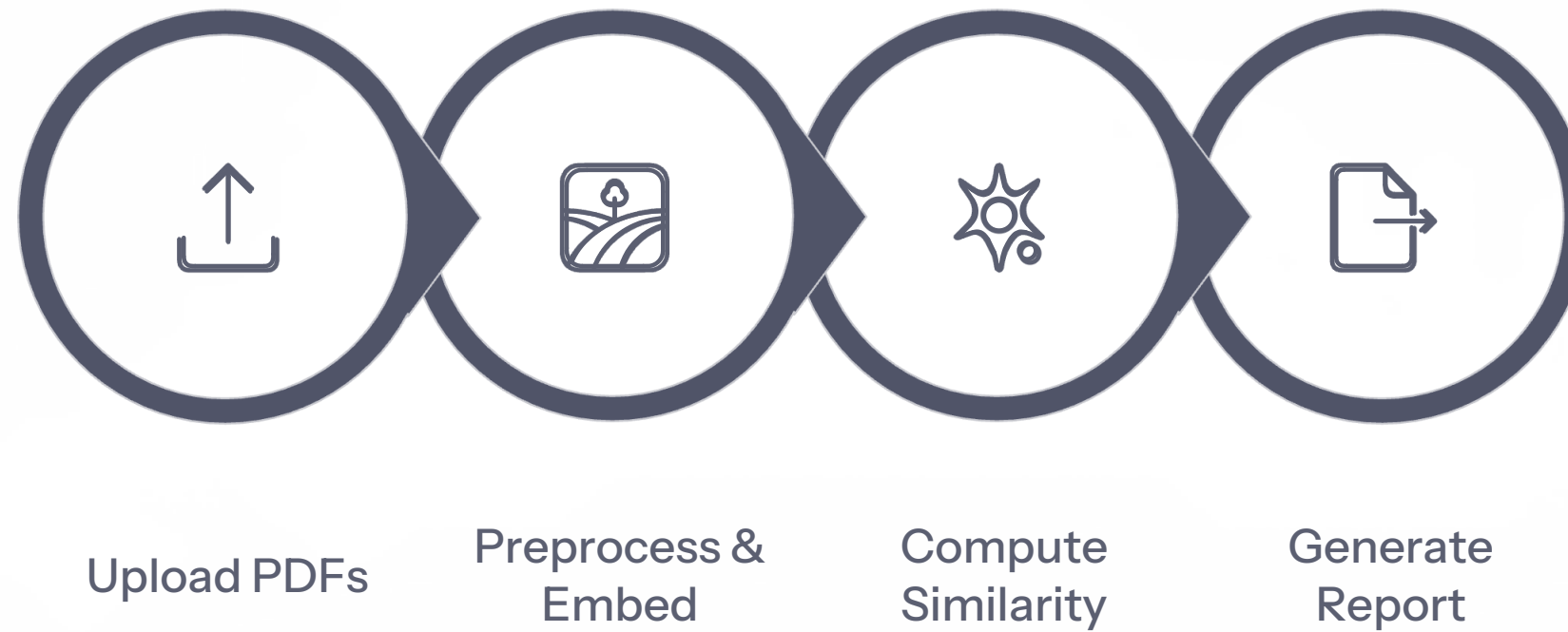
The system combines **Sentence-BERT** for dense vector embeddings with **cosine similarity** scoring to quantify semantic overlap between sentences.



Technology Stack

Python	Core language
Sentence-BERT	Semantic embeddings
PyPDF2	PDF text extraction
NLTK / spaCy	Text preprocessing
Scikit-learn	Cosine similarity
ReportLab	PDF report generation

Pipeline, Use Case & Activity Flow



The end-to-end pipeline transforms raw PDFs into an annotated plagiarism report in four stages — from ingestion through embedding, scoring, and final output.

Primary Use Cases

- **Academic institutions** — verify student submissions against external sources
- **Researchers** — check manuscript originality before publication
- **Content platforms** — detect duplicate or paraphrased articles at scale

Similarity Thresholds

> 0.80 Strong plagiarism	0.60–0.80 Possible paraphrase
< 0.60 Not plagiarism	

Models & Datasets

Core Model: Sentence-BERT

Sentence-BERT (SBERT) fine-tunes BERT with siamese networks to produce **semantically meaningful sentence embeddings**. Unlike vanilla BERT, SBERT enables direct cosine similarity comparison.

📄 SBERT reduces sentence comparison from $O(n^2)$ BERT inference to a single dot product — enabling real-time plagiarism detection.

Label 1

Paraphrase (plagiarism-like)

Dataset: PAWS

Paraphrase Adversaries from Word Scrambling — Google Research. Labeled sentence-pair dataset for binary classification.

Supporting: Wikipedia Sentences Dataset for custom plagiarism simulation.

Label 0

Not similar

Results

~75%

PAWS Accuracy

BERT-based classification on paraphrase detection benchmark.

0.80

Plagiarism Threshold

Scores above this indicate strong semantic plagiarism.

768

Embedding Dimensions

Each sentence encoded as a dense vector for comparison.

3

Output Tiers

Strong plagiarism, possible paraphrase, or not plagiarism.

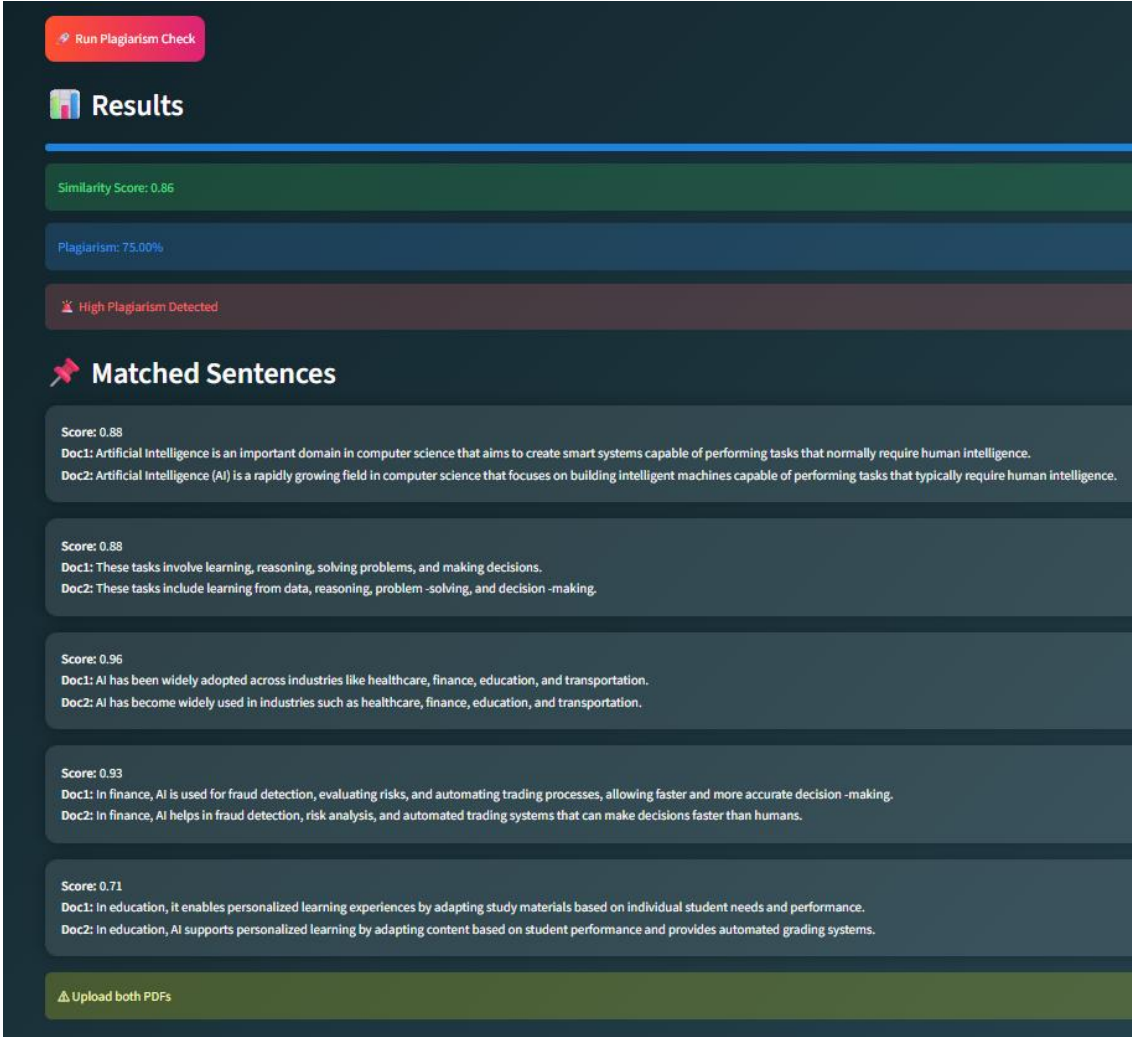
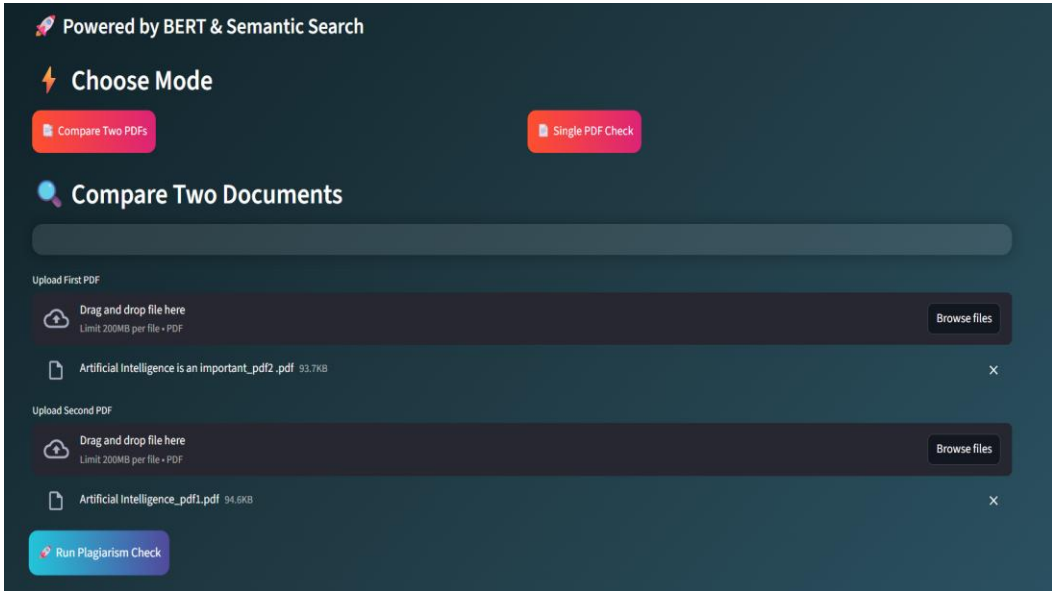
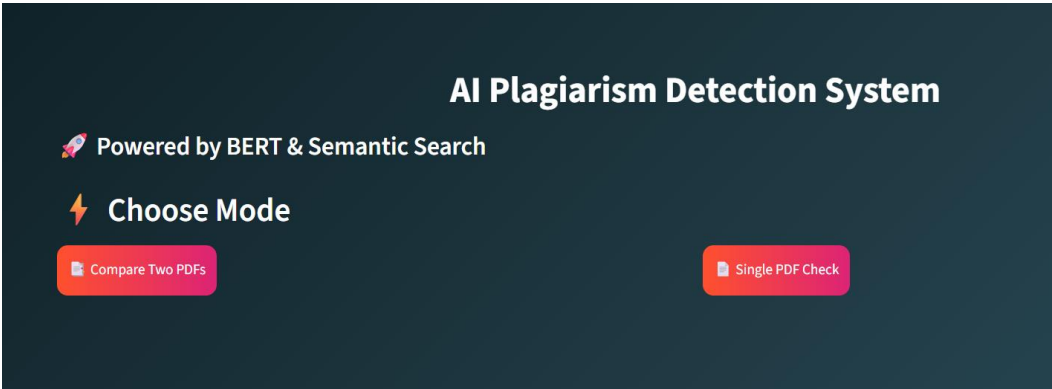
Performance Strengths

- High accuracy on **paraphrased and semantically similar** text
- Sentence-level granularity enables **precise highlighting**
- Embedding reuse reduces **inference latency**

Known Limitations

- Moderate accuracy on **completely rewritten** content
- Threshold tuning required per **domain and language**
- PDF extraction quality depends on **document formatting**

Snap Shots



Snap Shots

AI Plagiarism Detection System

🚀 Powered by BERT & Semantic Search

⚡ Choose Mode

📄 Compare Two PDFs

📄 Single PDF Check

📄 Single Document Detection

Upload PDF

📄 Drag and drop file here

Limit 200MB per file • PDF

Browse files

📄 Climate change_pdf3.pdf

94.1KB

×

🔍 Detect Plagiarism

📄 Result

Plagiarism: 0.00%

✅ Fully Original

📄 Highlighted Content

↔

Climate change is one of the most critical challenges facing the world today. It refers to long-term changes in temperature and weather patterns, mainly caused by human activities such as burning fossil fuels, deforestation, and industrial pollution. These activities increase greenhouse gas emissions, which trap heat in the atmosphere and lead to global warming. Climate change leads to rising global temperatures, melting glaciers, and extreme weather events like floods, droughts, hurricanes, and heatwaves. These changes have a serious impact on ecosystems, wildlife, and human life. Many species are at risk of extinction due to habitat loss and changing environmental conditions. Coastal areas are also threatened by rising sea levels, which can lead to displacement of communities. Governments and organizations around the world are working together to reduce carbon emissions and promote sustainable practices. One of the key solutions is the use of renewable energy sources such as solar, wind, and hydroelectric power. These energy sources are environmentally friendly and help reduce dependence on fossil fuels. Many countries are also implementing policies to control pollution and encourage green technologies. Individuals can also play an important role in combating climate change. Simple actions such as conserving energy, reducing waste, using public transport, and recycling can make a significant difference. Raising awareness about environmental issues is also essential to encourage collective action. Addressing climate change is crucial to ensure a sustainable future for coming generations. It requires global cooperation, technological innovation, and responsible behavior from both governments and individuals. If proper measures are taken, it is possible to reduce the impact of climate change and protect the planet for the future.

🔔 Detected Sentences

🔴 Climate change leads to rising global temperatures, melting glaciers, and extreme weather events like floods, droughts, hurricanes, and heatwaves.

Similarity: 0.61

THANK YOU